Parametric inference of Markov processes from partial observations

P. Chigansky

Le Mans, September 17-18, 2026

The Hebrew University of Jerusalem, Israel



During the postdoc in Le Mans, 2007

The postdoc project

- continuous time Markov chain X with k states and unknown transition rates $\theta = (\theta_{ij})$
- observed in "white noise"

$$Y_t = \int_0^t h(X_s)ds + W_t, \quad t \in \mathbb{R}_+$$

  where $h(\cdot)$ is known and W is the Bm, independent of X
- Find $T \to \infty$ asymptotics of MLE for θ, based on the sample

$$Y^T = \{Y_t, t \in [0, T]\}$$

Theorem (Ch. 2009)

Assume [...] then the MLE is AN at the rate $\sqrt{T}$ with the limit variance [...] and is asymptotically efficient.

Speaking more generally

- Markov state process X with generator $\mathcal{A}_\theta$
- The observed process

$$Y_t = \int_0^t h_\theta(X_s)ds + W_t$$

  where W is the Brownian motion, independent of X.
- Estimate θ given the realized path

$$Y^T = \{Y_t, t \in [0, T]\}.$$

Problem

Construct an asymptotically efficient estimator of θ in a relevant regime (large sample, small noise, etc.).

[**Q**]: why is it hard ?

The two forms of the likelihood

Two alternative forms for the likelihood function

$$L_T(\theta, Y^T) := \frac{d\mathsf{P}_{T,\theta}}{d\mathsf{P}_T}(Y^T), \quad \text{where } Y^T \sim P_{T,\theta},\ W^T \sim P_T$$

1. The mixture form (Kallianpur-Striebel formula):

$$L_T(\theta, Y^T) = \int_{C[0,T]} \exp\left(\int_0^T h_\theta(x_t)dY_t - \frac{1}{2}\int_0^T h_\theta(x_t)dt\right) d\mathsf{P}_\theta^X(dx)$$

The two forms of the likelihood

2. The filtered form:

$$L_T(\theta, Y^T) = \exp\left(\int_0^T \pi_t^\theta(h_\theta) dY_t - \frac{1}{2}\int_0^t \pi_t^\theta(h_\theta)^2 dt\right),$$

where the notation $\pi_t^\theta(f) := \mathsf{E}_\theta\big(f(X_t)|\mathcal{F}_t^Y\big)$ is used.

[**Q**]: How to compute the *filtering* projection $\pi_t^\theta(f)$?

[**A**]: The Kushner-Stratonovich filtering SPDE:

$$d\pi_t^\theta(f) = \pi_t^\theta(\mathcal{A}_\theta f)\, dt + \big(\pi_t^\theta(fh_\theta) - \pi_t^\theta(f)\pi_t^\theta(h)\big)\big(\underbrace{dY_t - \pi_t^\theta(h_\theta)\, dt}_{\overline{W}}\big), \quad \forall f$$

where $\overline{W}$ is the innovation Brownian motion $\mathcal{F}_t^Y = \mathcal{F}_t^{\overline{W}}$.

There are two finite dimensional cases

1. (X, Y) is Gauss-Markov
   - X is the OU process
   - Linear observation function $h_\theta(x) = h(\theta)x$, i.e.

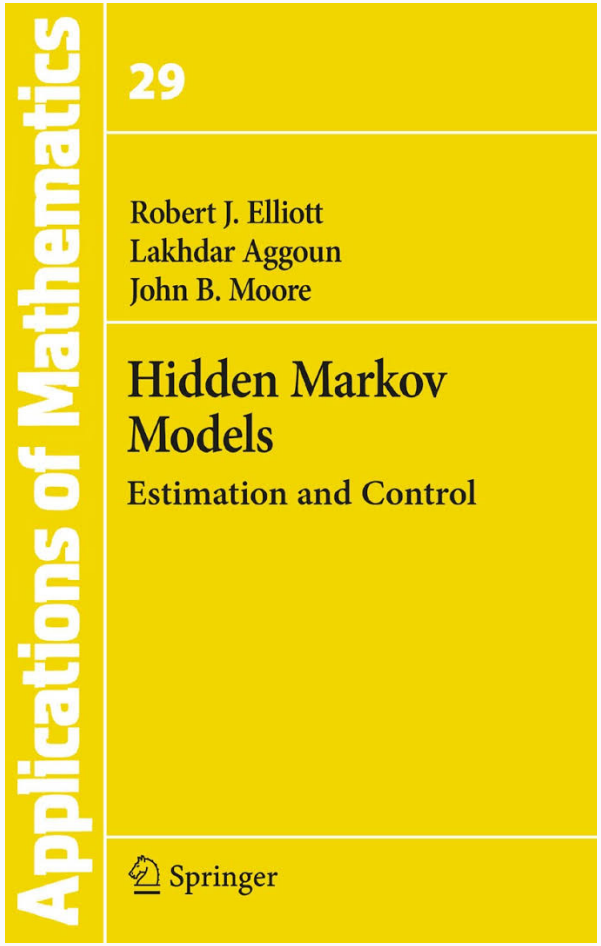
$$Y_t = h(\theta) \int_0^t X_s ds + W_t,$$

   - The solution of K-S SPDE reduces to the Kalman-Bucy (linear) SDE

2. X has finitely many states (the postdoc problem)
   - X is a Markov chain with k states
   - The K-S SPDE reduces to the Wonham (nonlinear) SDE on the simplex S^{k-1}

Methods for the finite dimensional cases

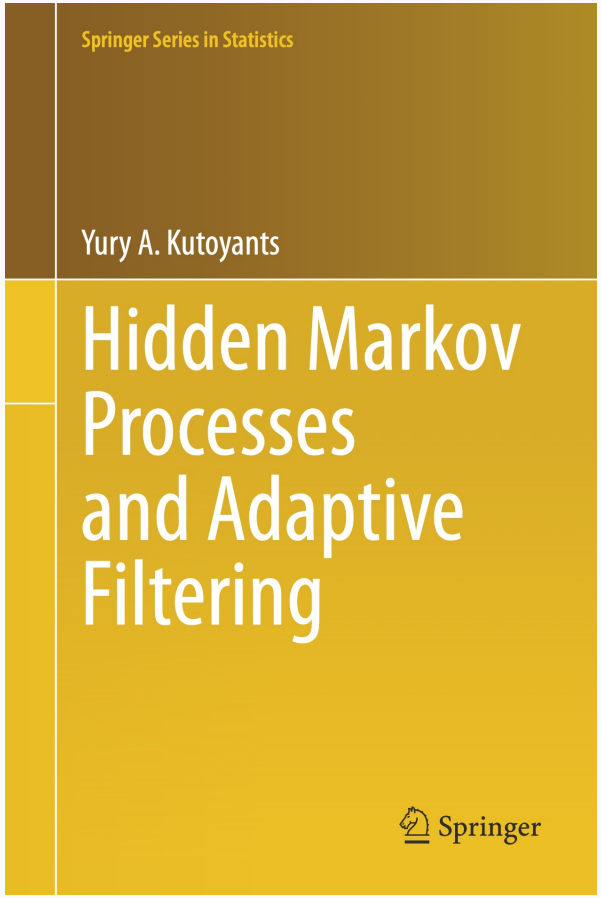
- Expectation-Minimization (EM) algorithm for MLE



(requires solving the filtering SDE on each iteration + usual troubles of EM)

Methods for the finite dimensional cases

- Fisher scoring method (One-Step MLE)



(no iterations, asymptotically efficient, non-generic preliminary estimator)

Theory in the finite dimensional cases

1. identifiability condition $(*)$
2. consistency
3. asymptotic normality $(*)$
4. asymptotic efficiency

$(*)$: explicit only when the state is single-variate

Why is it hard?

Methodology.

Iterative methods (optimization, integration, etc.) are infeasible:

- Generically, the S-K SPDE (and the unnormalized bilinear DMZ SPDE) are infinite dimensional (sometimes, provably so!)
- Numerical solvers tend to accumulate errors

Why is it hard?

Theory

1. Identifiability

$$\inf_{|\theta-\theta_0|>r} \mathsf{E}^{s}_{\theta_0} \left\| \pi^{\theta}_t(h_\theta) - \pi^{\theta_0}_t(h_{\theta_0}) \right\|^2 > 0, \quad \forall r > 0,$$

where $\mathsf{E}^{s}_{\theta_0}$ is w.r.t. (unique) invariant measure under P_{θ_0}.

2. The Fisher information matrix rate:

$$\frac{1}{T} \int_0^T \left(\nabla^\top \pi^{\theta}_t(h_\theta) \right) \left(\nabla \pi^{\theta}_t(h_\theta) \right) \xrightarrow[T\to\infty]{\mathsf{P}_\theta} I(\theta).$$

Why is it hard ?

Filtering K-S SPDE:

- stability ? exponential ? uniform ? global ?
- (unique) ergodicity ?
- smoothness in parameter variable ?
- derivatives ?
- limit theorems ?

Current theory typically requires that the state process X

1. takes values in a compact set
2. has "elliptic" transition operator

Thank you,

Prof. Kutoyants!